moPPIt for RBX1 binder design

Anthony Gitter[1,2,*]

[1]Department of Biostatistics and Medical Informatics, University of Wisconsin-Madison
[2]Morgridge Institute for Research
[*]gitter@biostat.wisc.edu

Introduction

The GEM x Adaptyv: RBX1 Binder Design Competition[1] tasked participants to submit binder sequences for the target protein RBX1[2]. RBX1 contains an N-terminal intrinsically disordered region (residues 1-39 in DisProt[3]), and submitted sequences had to be *de novo*. These elements added new challenges to this competition relative to the earlier EGFR Binder Design Competition[4]. To address these challenges and assess the potential to directly target the disordered region, I used the motif-specific PPI targeting algorithm (moPPIt)[5] for sequence design. moPPIt is a sequence-based binder design algorithm that combines a peptide-protein binding predictor built on top of ESM-2-650M[6] with Multi-Objective-Guided Discrete Flow Matching[7] for multi-objective sequence design. Part of the appeal is that it had previously successfully designed peptides that were validated to bind the N-terminal intrinsically disordered region of β-catenin[5].

Methods

My overall strategy was to use a pure moPPIt workflow. I prioritized the candidate binder sequences from moPPIt using moPPIt's generated metrics without considering other scores, such as Boltz-2-predicted interaction scores[8], to filter or rerank candidates. I first created a Docker image loosely derived from the moPPIt Colab notebook. The image contains the Python dependencies as well as the moPPIt code and weights from Hugging Face. Using a Docker image simplified running moPPIt in the University of Wisconsin-Madison Center for High Throughput computing[9] with HTCondor[10].

I used the complete RBX1 sequence from Proteinbase as the input target sequence, the motif penalty argument, 600 output sequences per run (n_batches), and all six objectives. I swept over all other moPPIt arguments in a grid search (Table 1). The motifs spanned all 20 residue long sliding windows along the sequence, including the intrinsically disordered region. Only one motif was provided per moPPIt run. The weight refers to the six objectives in this order: Hemolysis, Non-Fouling, Half-Life, Affinity, Motif, and Specificity. Overall, there were $4*9*5 = 180$ parameter combinations tested.

Parameter	Values
Sequence length	10, 15, 20, 25
Motif region	1-20, 11-30, 21-40, 31-50, 41-60, 51-70, 61-80, 71-90, 81-100
Weight combination	"1 1 1 4 4 2", "1 1 1 8 4 2", "1 1 1 8 2 1", "1 1 1 10 1 1", "1 1 1 4 8 4"

Table 1. moPPIt parameter ranges.

To rank the sequences, I first considered ranking based on the output affinity score or ranking by a weighted combination of affinity, motif recognition, and specificity. I ultimately decided to filter the candidate sequences, retaining only sequences that were unique, contained standard amino acids, had amino acid frequency of at most 4x that of UniProt sequences, had at most 3 identical amino acids in a row, had affinity score ≥ 8.0, had motif score ≥ 0.5, had specificity score ≥ 0.8, and had half-life score > 25.0. I ranked the remaining sequences by affinity score only and selected the top 100 to submit to the competition. All 100 were valid per the Proteinbase assessment.

Results

Each parameter combination ran independently in the Center for High Throughput computing on a single NVIDIA A100 80GB, L40, or L40S GPU. The runtimes ranged from 10h 29min to 20h 11min with a median of 13h 51min and a total of 105 GPU days (2,485h) of computing. The 180 moPPIt runs produced 600 peptide sequences each for a total of 108,000. Of these, 107,996 were unique. Applying the score filters removed most sequences. Only 856 remained. Applying the sequence composition filters further reduced the number of candidate sequences to 150. The score filter thresholds were chosen based on an inspection of the score distributions over all 108,000 sequences (Figure 1).

All of the 100 submitted sequences had length 20 or 25. The absence of length 10 or 15 sequences is primarily due to the half-life score filter. Half of the sequences targeted the intrinsically disordered region of RBX1 (Table 2).

Motif	Count
11-30	7
21-40	43
31-50	35
41-60	14
81-100	1

Table 2. Submitted sequence target motifs.

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Code availability

All analysis code and moPPIt outputs are available under the MIT license on GitHub: `https://github.com/agitter/adaptyv-rbx1`. The Docker image is available on DockerHub: `https://hub.docker.com/repository/docker/agitter/moppit`. I used Claude Sonnet 4.6[11] to draft most of the code.

References

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Appendix

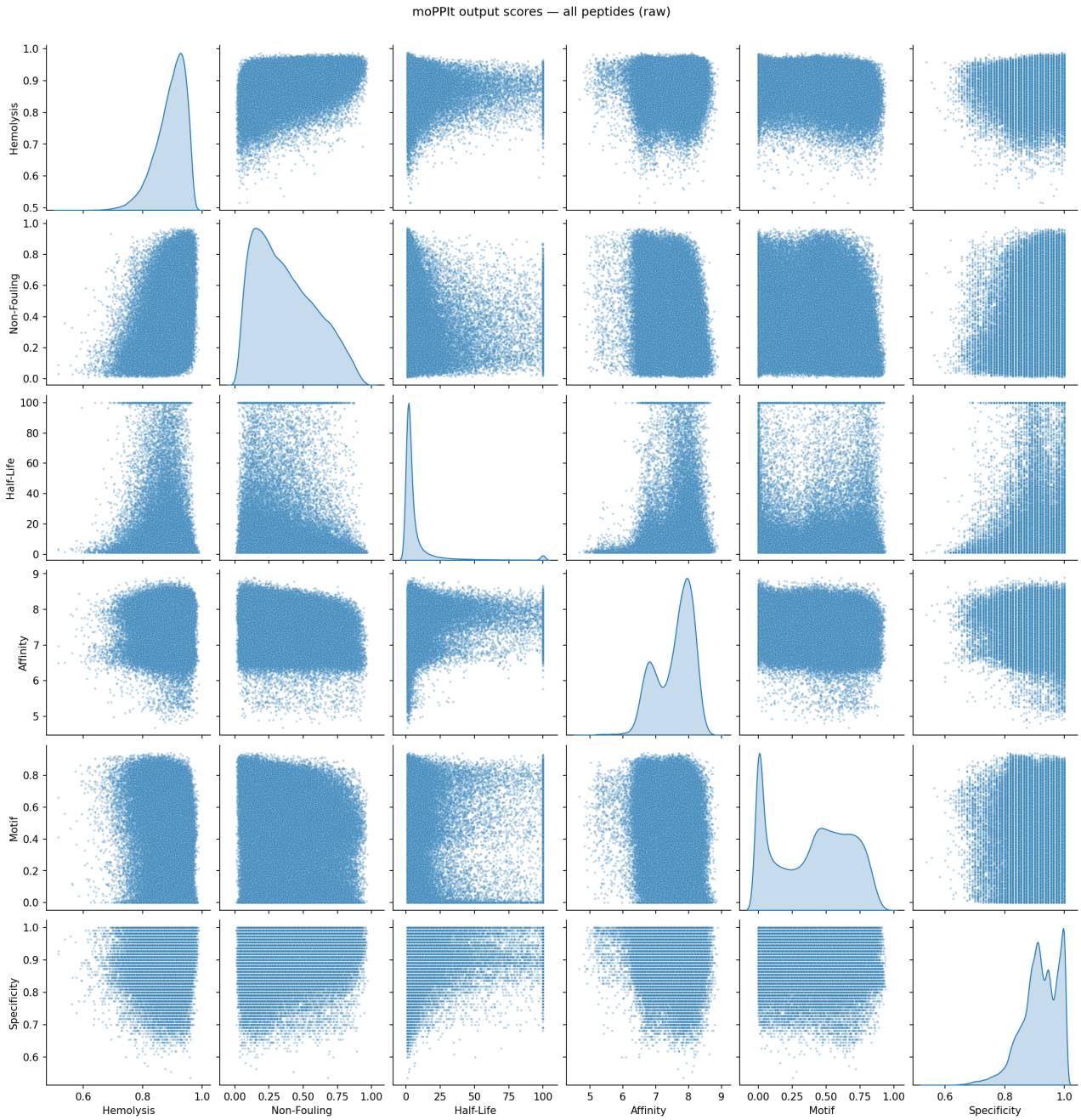

Figure 1. Pairplot of moPPIt scores for all 108,000 generated sequences.